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Education

- 2021 – 2026 **Doctor of Philosophy (Computer Science)**, *Nanyang Technological University*, Singapore
Advisor: Assoc. Prof. Han Yu
- 2005 – 2009 **B.S. in Mathematics and Applied Mathematics**, *Fuzhou University*, China

Research Interests

- Trustworthy Federated Learning
- Explainable Artificial Intelligence
- Generative Artificial Intelligence
- Graph Representation Learning

Work Experience

- Dec. 2025 – **Research Fellow**, *Bake AI*, USA
- Feb. 2026 I led the development of crust (**425+ GitHub stars**), an open-source security infrastructure for agentic AI systems that intercepts and blocks dangerous agent behaviors in real time.
- Jul. 2025 – **Research Fellow**, *Deeper Semantics Lab*, *NTU*, Singapore
- Nov. 2025
- Oct. 2019 – **Research Assistant**, *NTU-WeBank Research Centre on FinTech*, Singapore
- Jun. 2025 I contributed to multiple research projects at the Trustworthy Federated Ubiquitous Learning Research Centre.
- Jun. 2019 – **Research Intern**, *WeBank*, Shenzhen, China
- Sep. 2019 Federated learning is a popular distributed AI paradigm. During the internship, I implemented a federated-learning-based visual object detection system for operational safety, which received the AAAI 2020 Innovative Applications of Artificial Intelligence Award.
- May 2016 – **System Architect**, *DeepIR Inc*, Xiamen, China
- Feb. 2019 DeepIR delivers AI-powered services. My primary projects included an AI-based security module used at construction sites of the Beijing 2022 Winter Olympics venues, and a vehicle license plate recognition module used at the Hong Kong Customs checkpoint of the Hong Kong-Zhuhai-Macau Bridge.
- Jul. 2009 – **Senior Software Engineer**, *ChinaSource Ltd*, Xiamen, China
- Apr. 2016 Email service was one of the company's core businesses. Building on the open-source ecosystem, I developed a commercial mailing framework supporting the POP3, IMAP, and SMTP protocols.

Publications

Conference Proceedings

- Minghui Chen, Ruinan Jin, Wenlong Deng, **Yuanyuan Chen**, Zhi Huang, Han Yu, and Xiaoxiao Li, “Can Textual Gradient Work in Federated Learning?,” *ICLR* 2025.
- Chen Chen, Ruizhe Li, Yuchen Hu, **Yuanyuan Chen**, Chengwei Qin, and Qiang Zhang, “Overcoming Catastrophic Forgetting by Exemplar Selection in Task-oriented Dialogue System,” *ACL* 2024.
- Anran Li*, **Yuanyuan Chen***, Jian Zhang, Mingfei Cheng, Yihao Huang, Yueming Wu, Anh Tuan Luu, and Han Yu, “Historical Embedding-Guided Efficient Large-Scale Federated Graph Learning,” *SIGMOD* 2024.
- **Yuanyuan Chen***, Zichen Chen*, Pengcheng Wu, and Han Yu, “FedOBD: Opportunistic Block Dropout for Efficiently Training Large-scale Neural Networks through Federated Learning,” *IJCAI* 2023.

- **Yuanyuan Chen***, Zichen Chen*, Sheng Guo, Yansong Zhao, Zelei Liu, Pengcheng Wu, Chengyi Yang, Zengxiang Li, and Han Yu, “Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout,” *IAAI* 2023.
- Zelei Liu*, **Yuanyuan Chen***, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang, “Contribution-Aware Federated Learning for Smart Healthcare,” *IAAI* 2022.
- **Yuanyuan Chen**, Boyang Li, Han Yu, Pengcheng Wu, and Chunyan Miao, “HyDRA: Hypergradient Data Relevance Analysis for Interpreting Deep Neural Networks,” *AAAI* 2021.
- Yang Liu, Anbu Huang, Yun Luo, He Huang, Youzhi Liu, **Yuanyuan Chen**, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang, “FedVision: An Online Visual Object Detection Platform Powered by Federated Learning,” *IAAI* 2020.

Journal Articles

- Chuan Sun, **Yuanyuan Chen**, Xiaoli Tang, and Han Yu. 2026. “Efficient Federated Learning with Mean Block Difference-based Global Aggregation and Patience-based Local Training.” *IEEE Transactions on Knowledge & Data Engineering*, no. 01 (April): 1–16.
- Rui Liu, **Yuanyuan Chen**, Anran Li, Yi Ding, Han Yu, and Cuntai Guan. 2024. “Aggregating intrinsic information to enhance BCI performance through federated learning.” *Neural Networks* 172:106100.
- Zelei Liu*, **Yuanyuan Chen***, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang. 2023. “CAreFL: Enhancing smart healthcare with Contribution-Aware Federated Learning.” *AI Magazine* 44 (1): 4–15.
- Hao Wang, Boyang Li, Haoming Zhong, Ahong Xu, Yingjie Huang, Jingfu Zou, **Yuanyuan Chen**, et al. 2022. “Smart Decision-Support System for Pig Farming.” *Drones* 6 (12).
- Zelei Liu*, **Yuanyuan Chen***, Han Yu, Yang Liu, and Lizhen Cui. 2022. “GTG-Shapley: Efficient and Accurate Participant Contribution Evaluation in Federated Learning.” *ACM Transactions on Intelligent Systems and Technology* 13, no. 4 (May).
- Yang Liu, **Anbu Huang**, Yun Luo, He Huang, Youzhi Liu, Yuanyuan Chen, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang. 2021. “Federated Learning-Powered Visual Object Detection for Safety Monitoring.” *AI Magazine* 42, no. 2 (October): 19–27.

Awards & Honors

Academic Awards

- 2024 **Certificate of Accomplishment for your outstanding performance and dedication towards the ACM, Siggraph & Sigmobility NTU Student Chapter, NTU Association for Computing Machinery, Siggraph and Sigmobility**, Singapore
- 2023 **Deployed Application Award for “Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout”**, Association for the Advancement of Artificial Intelligence
- 2022 **Deployed Application Award for “Contribution-Aware Federated Learning for Smart Healthcare”**, Association for the Advancement of Artificial Intelligence
- 2020 **Innovative Applications of Artificial Intelligence Award for “FedVision: An Online Visual Object Detection Platform Powered by Federated Learning”**, Association for the Advancement of Artificial Intelligence

Industry Honors

- 2025 **NVIDIA FLARE Summer of Code 2025**, *NVIDIA FLARE*
- 2023 **PyTorch Community Awards for PyTorch Newcomer Nominee**, *The Linux Foundation*
Recognition for contributions to **PyTorch**, the world’s leading open-source deep learning framework powering modern AI research and industry deployments.
- 2021 **Mars 2020 Helicopter Contributor**, *GitHub*

- 2020 **Arctic Code Vault Contributor**, *GitHub*
University Awards
- 2024 **5 Years Long Service Award in appreciation for the service rendered to the university**, *Nanyang Technological University*, Singapore
- 2022 **Top 5 Most Popular Video Award for Trust-based Open Collaborative Federated Learning**, *College of Engineering, Nanyang Technological University*, Singapore

Research Grants

- 2021–2025 **Participant, “TrustFUL: Trustworthy Federated Ubiquitous Learning”**, *Singapore Research Programme Grant (No. AISG2-RP-2020-019)*, S\$6,883,200

Professional Services

Program Committee

- 2026 **The Fortieth AAAI Conference on Artificial Intelligence**, *AAAI-26*, Singapore
- Academic Reviewer
- 2026 The Fortieth AAAI Conference on Artificial Intelligence
- 2025 IEEE Transactions on Neural Networks and Learning Systems
- 2024 IEEE Transactions on Neural Networks and Learning Systems
- 2024 International Journal of Intelligent Systems
- 2023 ACM Transactions on Software Engineering and Methodology
- 2022 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences
- 2022 IEEE Transactions on Neural Networks and Learning Systems
- 2020 The Thirty-Fourth AAAI Conference on Artificial Intelligence
- Academic Conference Volunteer
- 2024 **The 2024 ACM Web Conference**, *WWW’24*, Singapore
- 2023 **The Thirty-Second International Joint Conference on Artificial Intelligence**, *IJCAI 2023*, Macao S.A.R.

Invited Talks & Presentations

- 2025 **NVIDIA FLARE Day APAC 2025**, *NVIDIA FLARE*, Online
- 2023 **Singapore ACM SIGKDD Symposium 2023**, *Singapore ACM SIGKDD Chapter*, Singapore
- 2018 **Invited Talk for Google Developer Groups**, *DevFest*, Xiamen, China

Projects

Award Winning Projects

- 2023 **FedOBD**
The Federated Opportunistic Block Dropout (FedOBD) approach was deployed for industrial fault diagnostic model training. It significantly reduces the communication overhead while maintaining model performance.
- 2022 **Contribution-Aware Federated Learning Framework**
The Contribution-Aware Federated Learning framework is designed and deployed for smart healthcare. It provides fair and explainable FL participant contribution evaluation in an efficient and privacy-preserving manner.
- 2020 **FedVision**
FedVision is a machine learning engineering platform to support the development of federated learning powered computer vision applications.

Other Projects

2017–2018 **AI Empowered Security System**

The system was deployed at the construction sites of the Beijing 2022 Winter Olympics venues.

2017 **Vehicle License Plate Recognition System for HZMB**

The system was developed in collaboration with partners for the Hong Kong–Zhuhai–Macau Bridge (HZMB) under the regulation and guidance of Hong Kong Customs.

2013–2016 **Commercial Mailing System**

A commercial mailing system was designed and developed on top of the open-source ecosystem. Common mailing protocols were supported as well as commercial extensions.

Membership

Professional Societies

2023–present **Member**, *Institute of Electrical and Electronics Engineers*

2022–present **Member**, *Association for Computing Machinery*

2022–present **Member**, *Association for the Advancement of Artificial Intelligence*

2016–2018 **Contributor**, *Association for Computing Machinery, Siggraph & Sigmobile NTU Student Chapter*

Engineering Organizations

2024–present **Approver**, *ONNX*

Approver in **SIG-archinfra** (Architecture & Infrastructure) — elevated from Contributor by SIG chair sponsorship per ONNX open governance.

2020–present **Outside Collaborator**, *PyTorch*

A privileged role granted by PyTorch maintainers to a select group of trusted external contributors, with code-review and issue-triage rights on core repositories. Sustained contributions since 2020 were recognised with the **PyTorch Community Award** (2023 Newcomer Nominee).

2024–present **Contributor**, *Protocol Buffers*

Protocol Buffers are language-neutral, platform-neutral extensible mechanisms for serializing structured data.

2024–present **Contributor**, *NVIDIA NVFlare*

NVIDIA FLARE is a domain-agnostic, open-source, and extensible SDK for Federated Learning.

2023–present **Contributor**, *Hugging Face*

The platform where the machine learning community collaborates on models, datasets, and applications.

2021–present **Contributor**, *NVIDIA DALI*

The NVIDIA Data Loading Library (DALI) is a GPU-accelerated library for data loading and pre-processing to accelerate deep learning applications.

2018–present **Contributor**, *OpenCV*

OpenCV is the world's biggest computer vision library.

2018–present **Contributor**, *Apache Thrift*

Thrift is a lightweight, language-independent software stack for point-to-point RPC implementation.

2018–present **Contributor**, *Faiss*

Faiss is a library for efficient similarity search and clustering of dense vectors.